King Fahd University of Petroleum and Minerals
Department of Mathematics
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Math 102 Major Exam II Term 241

Original exam date: 6 November 2024. Practice version: questions and choices shuffled.
Net time allowed: 90 minutes

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1. The area of the surface generated by revolving the curve $y = \frac{1}{3}x^3$ on the interval $[0, 3]$ about the x-axis is

(a) $\frac{\pi}{3}\left(26\sqrt{26} - 1\right)$

(b) $\frac{\pi}{3}\left(82\sqrt{82} - 1\right)$

(c) $\frac{\pi}{9}\left(82\sqrt{82} + 1\right)$

(d) $\frac{\pi}{9}\left(82\sqrt{82} - 1\right)$

(e) $\frac{\pi}{9}\left(26\sqrt{26} - 1\right)$

2. $\int \sec^6 4x \, \tan 4x \, dx =$

(a) $\frac{\sec^6 4x}{6} + c$

(b) $\frac{\sec^6 4x}{24} + c$

(c) $\frac{\sec^7 4x}{12} + c$

(d) $\frac{\sec^6 4x}{4} + c$

(e) $\frac{\sec^6 4x}{12} + c$

3. If $\frac{2x^3 - 4x - 8}{x(x-1)(x^2+4)} = \frac{2}{x} - \frac{2}{x-1} + \frac{Cx + D}{x^2 + 4}$, then $C + D =$

(a) 6

(b) 0

(c) 8

(d) 2

(e) 4

4. $\int_0^{\pi/2} \dfrac{d\theta}{3-2\cos\theta} =$

(a) $\dfrac{2}{\sqrt{5}} \arctan\sqrt{5}$

(b) $\dfrac{1}{\sqrt{5}} \arctan 5$

(c) $\sqrt{5} \arctan\sqrt{5}$

(d) $\dfrac{1}{\sqrt{5}} \arctan\sqrt{5}$

(e) $\dfrac{2}{\sqrt{5}} \arctan 5$

5. $\int \dfrac{1}{(16-x^2)^{3/2}}\, dx =$

(a) $\dfrac{1}{8}\left(\dfrac{x}{\sqrt{16-x^2}}\right) + c$

(b) $\dfrac{x}{\sqrt{16-x^2}} + c$

(c) $\dfrac{1}{4}\left(\dfrac{x}{\sqrt{16-x^2}}\right) + c$

(d) $\dfrac{1}{2}\left(\dfrac{x}{\sqrt{16-x^2}}\right) + c$

(e) $\dfrac{1}{16}\left(\dfrac{x}{\sqrt{16-x^2}}\right) + c$

6. $\int \dfrac{3-x}{3x^2-2x-1}\, dx =$

(a) $\ln|3x+1| - \ln|x-1| + c$

(b) $\dfrac{-5}{3}\ln|3x+1| + \dfrac{1}{2}\ln|x-1| + c$

(c) $\dfrac{-5}{3}\ln|3x+1| - \dfrac{1}{2}\ln|x-1| + c$

(d) $\dfrac{-5}{6}\ln|3x+1| - \dfrac{1}{2}\ln|x-1| + c$

(e) $\dfrac{-5}{6}\ln|3x+1| + \dfrac{1}{2}\ln|x-1| + c$

7. The arc length of the graph of the function $y = \frac{2}{3}(x^2+1)^{3/2}$ over the interval $[0,1]$ is

(a) $\frac{5}{3}$

(b) $\frac{2}{3}$

(c) $\frac{4}{3}$

(d) $\frac{4}{5}$

(e) $\frac{5}{4}$

8. The volume of the solid formed by revolving the region bounded by the graphs of $y = \sqrt{x}$, $y = 0$, $x = 1$ and $x = 4$, about the x-axis is:

(a) $\frac{17\pi}{2}$

(b) $\frac{9\pi}{2}$

(c) $\frac{11\pi}{2}$

(d) $\frac{15\pi}{2}$

(e) $\frac{13\pi}{2}$

9. $\int \frac{dx}{(x^2+1)^{3/2}}$

(a) $\frac{x}{\sqrt{x^2+1}} + c$

(b) $\frac{1}{\sqrt{x^2+1}} + c$

(c) $\frac{3x}{\sqrt{x^2+1}} + c$

(d) $\frac{\sqrt{x^2+1}}{x} + c$

(e) $\sqrt{x^2+1} + c$

10. $\int \dfrac{5x}{e^{2x}}\,dx =$

(a) $\dfrac{5}{4}e^{-2x}(2x+1)+c$

(b) $\dfrac{-5}{4}e^{-2x}(2x-1)+c$

(c) $\dfrac{-5}{4}e^{2x}(2x+1)+c$

(d) $\dfrac{-5}{4}e^{-2x}(2x+1)+c$

(e) $\dfrac{-5}{2}e^{-2x}(2x+1)+c$

11. $\int(\cot x)\ln(\sin x)\,dx =$

(a) $\ln(\sin x)+c$

(b) $[\ln(\cos x)]^2+c$

(c) $\dfrac{1}{2}[\ln(\sin x)]^2+c$

(d) $[\ln(\sin x)]^2+c$

(e) $\dfrac{1}{2}\ln(\sin x)+c$

12. $\int_0^{\pi/2}\cos^3 x\,dx =$

(a) $\dfrac{4}{3}$

(b) $\dfrac{5}{3}$

(c) $\dfrac{1}{3}$

(d) $\dfrac{2}{3}$

(e) 1

13. The volume of the solid formed by revolving the region bounded by the graphs of $y=x^2+1$, $y=0$, $x=0$, and $x=1$ about the y-axis is

(a) π

(b) $\dfrac{5\pi}{2}$

(c) 2π

(d) $\dfrac{\pi}{2}$

(e) $\dfrac{3\pi}{2}$

14. $\int 4 \arccos x \, dx =$

(a) $4x \arccos x + \sqrt{1 - x^2} + c$

(b) $4x \arccos x - 4\sqrt{1 - x^2} + c$

(c) $x \arccos x - \sqrt{1 - x^2} + c$

(d) $4 \arcsin x + c$

(e) $x \arccos x - 2\sqrt{1 - x^2} + c$

Answer Key

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Q	Ans	Brief explanation
1	**(d)**	$S = 2\pi \int_0^3 \frac{x^3}{3}\sqrt{1+x^4}\,dx = \frac{\pi}{9}(82\sqrt{82}-1)$.
2	**(b)**	$u = \sec 4x$: $\frac{1}{4}\int u^5\,du = \frac{\sec^6 4x}{24} + c$.
3	**(a)**	Comparing numerators gives $Cx + D = 2x + 4$, so $C + D = 6$.
4	**(a)**	$u = \tan\frac{\theta}{2}$: $\int_0^1 \frac{2\,du}{1+5u^2} = \frac{2}{\sqrt{5}}\arctan\sqrt{5}$.
5	**(e)**	$x = 4\sin\theta$: $\frac{1}{16}\int \sec^2\theta\,d\theta = \frac{1}{16}\frac{x}{\sqrt{16-x^2}} + c$.
6	**(e)**	$\frac{3-x}{(3x+1)(x-1)} = \frac{-5/2}{3x+1} + \frac{1/2}{x-1}$, giving $-\frac{5}{6}\ln\lvert 3x+1\rvert + \frac{1}{2}\ln\lvert x-1\rvert$.
7	**(a)**	$1 + (y')^2 = (2x^2+1)^2$, so $L = \int_0^1 (2x^2+1)\,dx = \frac{5}{3}$.
8	**(d)**	Disks: $V = \pi\int_1^4 x\,dx = \frac{15\pi}{2}$.
9	**(a)**	$x = \tan\theta$: $\int \cos\theta\,d\theta = \sin\theta = \frac{x}{\sqrt{x^2+1}} + c$.
10	**(d)**	By parts, $u = x$, $dv = e^{-2x}dx$: $-\frac{5}{4}e^{-2x}(2x+1) + c$.
11	**(c)**	$u = \ln(\sin x)$, $du = \cot x\,dx$: $\frac{1}{2}u^2 + c$.
12	**(d)**	$\int_0^1 (1-u^2)\,du = \frac{2}{3}$ with $u = \sin x$.
13	**(e)**	Shells: $V = 2\pi\int_0^1 x(x^2+1)\,dx = \frac{3\pi}{2}$.
14	**(b)**	By parts: $\int \arccos x\,dx = x\arccos x - \sqrt{1-x^2}$; times 4.